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A method for skin lesion detection and localization by means of Deep Learning and reliable prediction explainability

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ABSTRACT

Skin lesions are any abnormal growths or appearances on the skin, ranging from benign (i.e., non-cancerous) to malignant (i.e., cancerous). The identification of a skin lesion is a crucial task that is carried out in short periods of time to initiate an eventual therapeutic treatment. In this paper, we propose a method for automatic skin lesion detection, implementing Convolutional Neural Networks. Moreover, with the aim of providing a rationale behind the model prediction, we also consider explainability by adopting two different Class Activation Mapping algorithms, which highlight regions in skin images that most contribute to the network's classification decision. We also include the indices of similarity for further quantitative analysis. Several Convolutional Neural Networks are considered, by obtaining the best results with the MobileNet model, achieving an accuracy equal to 0.935 in skin lesion detection. Moreover, in the experimental analysis, we discuss the effectiveness of Class Activation Mapping algorithms exploited for skin lesion localization.

1. Introduction

Skin lesions are a diverse group of skin abnormalities that can range from benign conditions to potentially malignant cancer [1]. Skin cancer represents the most common cancer in the United States, affecting one out of every five people. It affects around 9,500 people per day, and its frequency increased by 145% between 1976 and 1984 and from 2000 to 2010. Melanoma is the seventh most common cancer diagnosis for both men and women in 2024. For both sexes, invasive melanoma is expected to be the sixth most common cancer diagnosed. Melanoma rates have continuously increased over the last 30 years, with a 31.5% increase from 2011 to 2019. The incidence of invasive melanoma has increased by 3% per year in women aged 50 and up, while it has stabilized in younger women.[1] In addition, over 100 thousand cases of melanoma skin cancer were registered in the European Union in 2020, while over 16 thousand deaths were recorded across the continent that same year. Northern European countries such as Denmark, the Netherlands, and Norway are among the countries with the highest rates of skin cancer in the world.[2]

An effective treatment for skin cancer depends on early detection, a global health concern. Dermatologists are skilled in visualizing skin lesions; yet, there are often difficulties due to the intricacy and minute differences between various types of lesions. As a matter of fact, the traditional diagnostic process, reliant on visual examination and sometimes biopsy, is time-consuming and requires substantial expertise. With the advent of advanced imaging technologies and the increasing prevalence of digital medical records, there is a growing opportunity to leverage computational methods to assist dermatologists in diagnosing skin conditions more efficiently and accurately [2].

Deep Learning (DL) [3] is a subfield of Machine Learning (ML), it can automatically learn and extract relevant features from unprocessed data, making it a valuable tool for image analysis [4–6]. Particularly, convolutional neural networks (CNNs) have shown interesting performance in a variety of image recognition applications, such as medical

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[1] https://www.aad.org/media/stats-skin-cancer.
[2] https://www.statista.com/topics/11101/skin-cancer-in-europe/topicOverview

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bioimaging analysis [6,7]. These networks are usually employed for identifying and classifying skin diseases from dermatoscopy images and recognizing recurring patterns [8].

Typically, CNN models are often considered black boxes, making it challenging for clinicians to interpret predictions. This feature presents serious problems, especially in delicate domains such as healthcare and in particular during the diagnosis procedure.

Explainability in CNNs refers to techniques exploited to make the model's decision-making process transparent and interpretable. In the context of medical imaging, explainability is crucial not only for gaining clinical acceptance but also for ensuring the reliability and safety of automated diagnostic tools. Clinicians and medical staff need to trust that the model predictions are based on medically relevant features and not on spurious correlations or data artifacts [9].

Starting from these considerations, in this paper we propose a method aimed at automatically detecting and localizing skin lesions. We use DL, with particular regard to CNNs and we propose several techniques aimed at integrating prediction explainability, including heatmaps provided by two different Class Activation Mapping (CAM) algorithms i.e., Gradient-weighted Class Activation Mapping (Grad-CAM [10]) and Score-Weighted Class Activation Mapping (Score-CAM [11]), to provide a comprehensive understanding of the model's decision-making process. The idea behind CAM applications is to provide visual explanations by displaying heatmaps of the class-discriminative regions. The Grad-CAM and Score-CAM algorithms are used to highlight areas of interest in a model. Both Grad-CAM and Score-CAM are valuable algorithms for enhancing the explainability of CNNs. Grad-CAM provides a straightforward method to visualize model decisions based on gradients, while Score-CAM offers improved accuracy and resolution by using score-based weighting.

To increase model explainability, Structural Similarity Index Measure (SSIM [12,13]) and Feature Similarity Index Measure (FSIM [14,15]) indices are computed. SSIM takes into account contrast, brightness, and distortions, while FSIM uses phase congruency and gradient magnitude to extract features. Both indices have specific range values, allowing comparisons between different models.

Thus, we aim to create a classification approach system that is reliable and accurate while also offering an understanding of the logic behind the model. The main goal is to develop a trustworthy and transparent tool that will help dermatologists with diagnosis by focusing on the areas of dermatoscopy images under analysis with visual explainability algorithms [16].

From a medical point of view, the relevance of explainable DL in the diagnosis process is highlighted by the importance of justifying the DL detection and localizing the areas of the image responsible for classification, improving credibility and trustworthiness [17,18]

Below we itemize the main contributions of the paper:

- we propose an explainable method for skin lesion detection;
- we consider a set of 5 different CNN;
- we take into account two different CAM techniques with the aim to provide reliability behind model predictions;
- we consider 7 different skin lesions i.e., Melanoma, Melanocytic nevus, Basal cell carcinoma, Actinic keratosis/Bowen's disease, Benign keratosis, Dermatofibroma, Vascular lesion i.e., we propose a seven class classification of skin images;
- in the evaluation of a real-world dataset composed of 38.569 skin lesion images we obtain (with the model obtaining the best performances), an accuracy equal to 0.936, a precision equal to 0.937 and a recall equal to 0.935;
- the two CAM evaluations confirm the interesting performance of the model during classification, highlight the same areas, linked to the model's recurrent patterns and the localization of the lesion in the images, improving reliability and trustworthiness;
- finally, the indices computation (i.e., FSIM and SSIM) are able to provide information on the quality of the predictions in terms of generalization and represent an additional confirmation of the good classification performances obtained.

The paper proceeds as follows: in the next section a review of the state-of-the-art of skin lesions DL detection; in Section 3 we present the proposed method for the explainable detection of skin lesions from images obtained from dermatoscopy; in Section 4 we exploit the results and discuss them; and, finally, in the last section, we present the conclusion and future research about this topic.

2. Related work

In the following section, we provide an overview of current state-of-the-art literature related the adoption of DL in the context of skin lesion classification.

For instance, Di Biasi et al. [19] introduces GACNN, a novel approach combining Genetic Algorithms (GA) and CNN for melanoma detection. Using the MED-NODE clinical dataset, the researchers classified melanoma and compared GACNN's performance against AlexNet, both with and without Otsu segmentation. Accuracy served as the scoring function for GA. Preliminary results indicate that GACNN has the potential to enhance melanoma classification by enabling automated network design, and the approach can be expanded to other melanoma datasets and optimization algorithms in the future. In evaluating melanoma classification performance, AlexNet achieved a maximum accuracy (ACC) of 0.97%, with an average ACC of 0.81% over 100 training repetitions. The GACNN system, evolving for 100 iterations, also reached a maximum ACC of 0.97% before the 100th iteration. Another interesting work of this group [2] addressed the need for early melanoma detection by developing Computer-Aided Diagnosis (CAD) systems that prioritize minimizing the false negative rate (FNR). To achieve a low FNR, the researchers conducted an analysis of various CNN architectures, adapting them for the melanoma clinical image binary classification problem. The goal was to identify the CNN architecture that performed best in terms of FNR. For this, they used the MED-NODE clinical image dataset, comprising 170 images (70 melanoma, 100 naevi). The CNNs optimized for this problem included AlexNet, DenseNet, GoogleNet Inception V3, GoogleNet, MobileNet, ShuffleNet, SqueezeNet, and VGG16. VGG and AlexNet were most effective in minimizing the FNR for melanoma detection, achieving FNRs of 0.07 and 0.13 respectively. While prioritizing a low FNR, both models maintained respectable global performance: VGG showed 73% accuracy, 82% sensitivity, and 59% specificity, and AlexNet achieved 89% accuracy, 87% sensitivity, and 90% specificity.

Using the HAM10000 dataset, the authors of [20] employed a set of ML models, including SVM, Random Forest, KNN, Decision Tree, and Ridge Classifier, to categorize the different types of skin cancers. Following a comprehensive evaluation, their models produced positive results. Compared to the other models studied, the Random Forest algorithm performs better, achieving 95.6% accuracy on the test set and 95.5% accuracy on the validation set. A fascinating study has been reported in [21]. To diagnose skin cancers like melanoma (MEL), basal cell carcinoma (BCC), squamous cell carcinoma (SCC), and melanocytic nevi (MN), the authors planned to develop a multi-classification technique. In their study, they presented a novel model, a CNN-based DL-based skin cancer classification network (DSCC_Net). They assessed it using three publicly accessible benchmark datasets (i.e., ISIC 2020, HAM10000, and DermIS). To diagnose skin cancer, the suggested DSCC_Net model's classification performance is evaluated against six baseline deep networks: ResNet-152, Vgg-16, Vgg-19, Inception-V3, EfficientNet-B0, and MobileNet. To address the minority classes problem in this dataset, they also employed SMOTE Tomek. The suggested DSCC_Net was able to classify the four different forms of skin cancer with an AUC of 99.43%, accuracy of 94.17%, recall of 93.76%,

precision of 94.28%, and F1-score of 93.93%. The work of Gouda et al. [22] focused on detecting malignant and benign skin tumors using DL methods on the ISIC2018 dataset (3533 skin lesions). Before training, ESRGAN (Enhanced Super-Resolution Generative Adversarial Networks) was uniquely applied for photo retouching and improvement, followed by standard preprocessing steps like augmentation, normalization, and resizing. The researchers also explored transfer learning models including Resnet50, InceptionV3, and Inception Resnet for fine-tuning and comparison against their custom-designed CNN model. The CNN model achieved an 83.2% accuracy rate. For comparison, the Resnet50 model reached 83.7%, InceptionV3 performed best at 85.8%, and Inception Resnet yielded 84% accuracy. The custom-designed CNN demonstrated performance comparable to the pre-trained models.

Moreover, the study of Guizar et al. [23] focused on the challenge of accurate skin lesion diagnosis by comparing U-Net and attention-based methods for skin lesion image segmentation, aiming to overcome limitations of traditional CNNs in capturing long-range spatial relations and pure Vision Transformers (ViT) with small medical datasets. The method involved a comprehensive comparative analysis of various approaches, including hybrid models combining ViT with ConvNets. This aimed to improve detailed localization information recovery crucial for skin lesion classification. Results indicate that the hybrid TransUNet model significantly outperformed other benchmarking methods, achieving an accuracy of 92.11% and a Dice coefficient of 89.84%. Researchers in [24] proposed a Swin Transformer model that synergistically combines the strengths of both Transformer and CNN architectures for end-to-end mapping. To tackle class imbalance, a weighted cross-entropy loss function is integrated. Their approach also details explorations of key components for efficient learning with multiclass data. Evaluated extensively on the ISIC 2019 Skin Lesion Analysis Towards Melanoma Detection Challenge dataset. The proposed method achieved a sensitivity of 82.3%, specificity of 97.9%, accuracy of 97.2%, and a balanced accuracy of 82.3%.

A work that implements XAI is the work of Ahmed et al. [25] . This work introduced a novel framework for skin lesion recognition, addressing challenges like lesion irregularities and low-contrast. Their methodology involved initial data augmentation to expand dataset size. Subsequently, two fine-tuned, pretrained DL models, Xception and ShuffleNet, extract deep features using a global average pooling layer. These selected features are then classified using ML algorithms. Furthermore, Grad-CAM-based visualization is employed to identify important image regions. The results demonstrate significant accuracy improvements on public datasets: ISIC2018 achieved 99.3% accuracy, and HAM10000 reached 91.5%.

A recent paper is reported in [26]. This work introduced SkinSage XAI, a framework for skin lesion classification that prioritizes interpretability. Their methodology addressed the "black-box" limitation of DL models by integrating XAI techniques. A diverse dataset of approximately 50,000 images from the Customized HAM10000 serves as the foundation. The Inception v3 model is employed for classifying seven types of skin lesions. To provide clear visual explanations for model outputs, the framework incorporates Grad-CAM and local interpretable model-agnostic explanations algorithms. The results demonstrate high performance, with SkinSage XAI achieving an accuracy of 96%. Other key metrics include a precision of 96.42%, recall of 96.28%, F1 score of 96.14%, and an area under the curve of 99.83%.

Finally, the schematization of the related work is reported in Table 1, which examines the authors, years, the method used, the quantitative results, and the adoption of XAI.

3. The method

This section details the proposed approach due to the explainable detection and localization of skin lesions. Fig. 1 shows the proposed method workflow.

3.1. Dataset

Starting from the fundamental step of the dataset's choice, the dermatoscopy images coming from the public dataset on the Kaggle website,[3] that is a pre-processed version of the HAM10000 dataset, which the original challenge is reported in the following link[4]: [27]. The lesion images come from the HAM10000 Dataset and were acquired with a variety of dermatoscopy types, from all anatomic sites, from a historical sample of patients presented for skin cancer screening, from several different institutions. Images were collected with the approval of the Ethics Review Committee of the University of Queensland and the Medical University of Vienna. The HAM10000 dataset is primarily composed of dermatoscopy images from individuals with fair skin, a known limitation that could potentially impact the model's generalizability and equitable performance across diverse ethnic groups and skin tones in real-world clinical applications.

The skin images are obtained by exploiting dermoscopy, also known as dermatoscopy i.e., a non-invasive imaging technique used to examine skin lesions and other skin abnormalities in detail. It enhances the visualization of skin structures that are not visible to the naked eye, making it an essential tool in dermatology for diagnosing various skin conditions, including skin cancer. Dermoscopy magnifies skin lesions, allowing the evaluation of pigmentation, vascular architecture, and surface patterns. It detects early indicators of skin cancer, notably melanoma, by examining unique patterns and changes in skin lesions. Dermoscopy is a technique that uses a specific instrument with a magnifying lens to enlarge a picture of the skin surface by 10- to 20 times. The device frequently has built-in illumination to highlight the skin, increasing the visibility of subsurface structures.

The seven classes (with labels) [27] to perform the classification through CNNs are listed below:

- MEL: "Melanoma" diagnosis. One type of skin cancer that begins in the melanocytes is called melanoma. The cells called melanocytes produce the pigment that gives skin its color. We refer to the pigment as melanin. Usually, melanoma begins on skin that is frequently exposed to sunlight. This covers the skin of the legs, arms, back, and face. Moreover, melanoma can develop in the eyes. In rare instances, it may occur internally, including in the throat or nose.
- NV: "Melanocytic nevus" diagnosis. A mole is called a melanocytic nevus in medical terminology. Nevi can show up on the body at any location. Since they are benign (not malignant), they usually do not need to be treated. Melanomas can form inside of melanocytic nevi in a very tiny percentage of cases. It is noteworthy that most cutaneous melanomas originate in skin that seems normal.
- BCC: "Basal cell carcinoma" diagnosis. One kind of skin cancer is called basal cell carcinoma. The skin's basal cells, which generate new skin cells as older ones age, are the source of basal cell carcinoma. Though it can take numerous forms, basal cell carcinoma typically manifests as a little translucent lump on the skin. The head and neck are two areas of the skin that are most commonly affected by basal cell carcinoma due to sun exposure.
- AKIEC: "Actinic keratosis/Bowen's disease (intraepithelial carcinoma)" diagnosis. Actinic keratosis (AK) and Bowen's disease are both precancerous skin conditions caused primarily by excessive sun exposure. They represent different stages in the development of squamous cell carcinoma (SCC), a type of skin cancer.
- BKL: "Benign keratosis" diagnosis. Seborrheic keratosis is a frequent benign (noncancerous) skin development. People tend to have more of them as they age. Seborrheic keratoses are typically

[3] https://www.kaggle.com/datasets/artakusuma/basedir/data
[4] https://challenge2018.isic-archive.com

Table 1
State-of-the-art of the skin lesions through dermatoscopy images DL classification.

Authors	Model	Result	Explainbility
Di Blasi et al. (2021)	GACNN	97.0% accuracy	no
Di Blasi et al. (2023)	VGG and AlexNet	VGG: 0.07 FNR and 73% accuracy AlexNet: 0.13 FNR and 89% accuracy	no
Velaga et al. (2023)	SVM, Random Forest, KNN, Decision Tree, and Ridge Classifier	95.6% accuracy	no
Tahir et al. (2023)	DSCC_Net	94.17% accuracy	no
Gouda et al. (2022)	CNN model based on ERSGAN	83.2% accuracy	no
Guizar et al. (2022)	hybrid TransUNet	92.11% accuracy	no
Ayas et al. (2023)	Swin Transformer with CNN	97.2% accuracy	no
Ahmed et al. (2023)	Xception and ShuffleNet	ISIC2018: 99.3% accuracy HAM10000: 91.5% accuracy	yes
Munjal et al. (2024)	SkinSage XAI	96% accuracy	yes

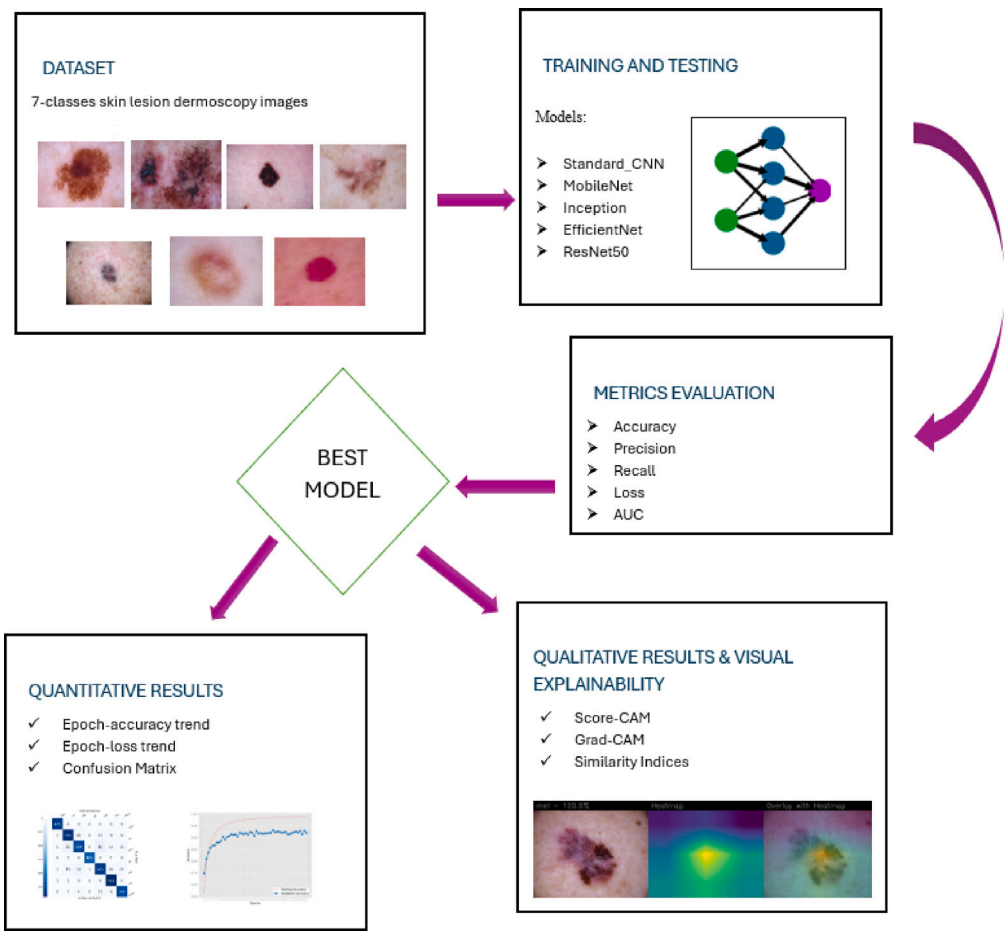

Fig. 1. The workflow of the proposed method for explainable skin lesion detection and localization.

brown, black, or light tan. The growths (lesions) seem waxy, scaly, and slightly elevated. They typically appear gradually, on the face, neck, chest, or back.

- DF: "Dermatofibroma" diagnosis. Dermatofibromas are tiny, innocuous growths on the skin. These growths, called papules, can appear anywhere on the body, but they are most frequent on the arms, lower legs, and upper back.
- VASC: "Vascular lesion" diagnosis. Vascular lesions are skin conditions caused by abnormalities in blood vessels. They can appear in various forms and colors, from red to purple or brown.

Fig. 2 shows samples of each skin lesion class under analysis.

second dataset that reports the same classes is used to the validation of the quantitative results for the CNNs.

3.2. The proposed approach: CNNs networks and visual explainability

The second stage is to train and test CNNs. All of the training and testing steps, as well as the hyperparameter setting and sample splitting, resulted in a network model (for each architecture) that was used to evaluate the outcomes. The CNNs applied are the following:

- *Inception*: The Inception [28] network is an important milestone in developing CNN classifiers. Before its inception (pun intended), most popular CNNs just stacked convolution layers deeper and deeper, hoping to get better performance;
- *ResNet50*: ResNet-50 [29], is a DL network comprising 50 layers. The network has an image input size of 224 × 224;

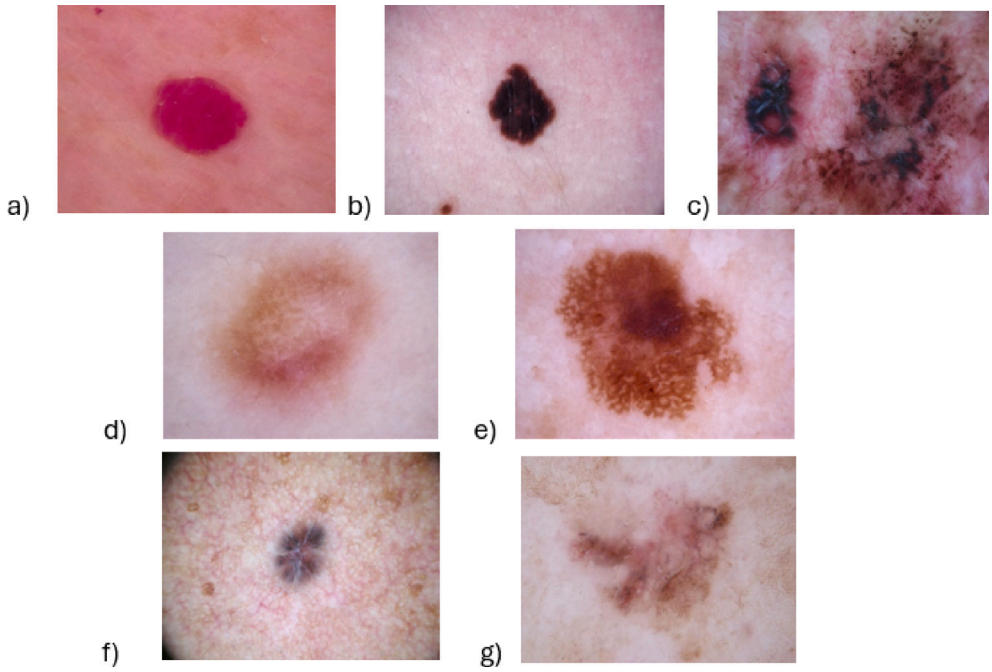

Fig. 2. Skin lesion samples: (a) vasc, (b) nv, (c) mel, (d) df, (e) bkl, (f) bcc, (g) akiac.

- *EfficientNet*: EfficientNet [30] is a DL network design and scaling method that uses a compound coefficient to uniformly scale all dimensions of depth, width, and resolution.
- *MobileNet*: MobileNet [31] employs depthwise separable convolutions. It considerably reduces the number of parameters when compared to a network with standard convolutions and the same depth in nets. This yields lightweight DL networks. Optional shape tuple that can only be supplied if include_top is False. Otherwise, the input shape must be (224, 224, 3).
- *Standard_CNN*: this network [32,33] is designed by the authors, It consists of a combination of 14 layers: Conv2D, MaxPooling, Flatten, Dropout and Dense. More information about this network can be found in [15].

Once trained and tested the networks in different hyper-parameters settings, and the quantitative best model is highlighted for further analysis. This evidence is related to the metrics results (i.e., accuracy, precision, recall, loss). Further analysis concerns the quantitative and qualitative points of view. Measured results consist of the confusion matrix and the graph representing the accuracy-epoch and the loss-epoch trends, useful to have a better overview of the best model performance. Finally, the correlation matrix on the metrics of the best output model provides correlation information over the independence contest of the seven skin lesion classes.

The next step, reported in Fig. 1, is related to the generation of activation maps using two distinct CAM algorithms. The goal is to provide visual explainability in terms of the localization of the skin lesions' common pattern in the dermatoscopy images. CAM-based algorithms make it possible to highlight the portions of the image that have contributed most to the classification, allowing us to understand the feature of the problem that is most discriminating, as well as highlighting possible regions not yet taken into account in current studies and thus guiding future developments. Moreover, the heatmaps are based on the VIRIDIS coloration,[5] regarding the pattern importance during the classification process. The Grad-CAM algorithm exploits the back-propagation of individual class weights to highlight areas of interest, while the Score-CAM algorithm solves the gradient saturation and false-confidence problems of Grad-CAM using a different approach. Score-CAM uses a parameter defined as Channel-wise Increase of Confidence (CIC) to assess the contribution of each feature map based on the class score.

Grad-CAM and Score-CAM are powerful techniques for visualizing and interpreting CNN model predictions. They provide insights into which parts of an image are most influential in determining the model's output, aiding in model validation, debugging, and improving transparency. While Grad-CAM relies on gradients and is straightforward to implement, Score-CAM offers higher-resolution and potentially more stable visualizations by using perturbation-based methods. Both methods are valuable for enhancing the prediction explainability of DL models across various applications, particularly in fields requiring high levels of accuracy and trust, such as medical imaging considered by the proposed method for skin lesion detection.

When both Grad-CAM and Score-CAM highlight the same area in a neural network's output, it signifies an important result for several reasons, such as the validation of model interpretability. The agreement between Grad-CAM and Score-CAM suggests that the identified region is genuinely important for the model's decision, thereby enhancing confidence in the interpretability of the model. Another important feature regards the robustness of the explanation. The consensus between different methods indicates that the explanation is robust and not merely an artifact of the assumptions or implementation of a particular technique. In addition, an agreement can provide deeper insight into which features or parts of the input the model considers important, aiding in understanding model behavior.

Moreover, to further increase the level of explainability of the models, we compute the SSIM and the FSIM indices. SSIM is an index that takes into account variations in contrast, brightness, and possible distortions relative to two versions of the same image. This index indicates how different two heatmaps produced by different CAM algorithms, using the same model, are. The FSIM extracts features from the input images and then calculates their similarities. It combines two digital image processing concepts: phase congruency, which extracts image information in the frequency domain, and gradient magnitude, which uses convolutional filters (or masks) to evaluate the gradients of the

[5] https://cran.r-project.org/web/packages/viridis/vignettes/intro-to-viridis.html

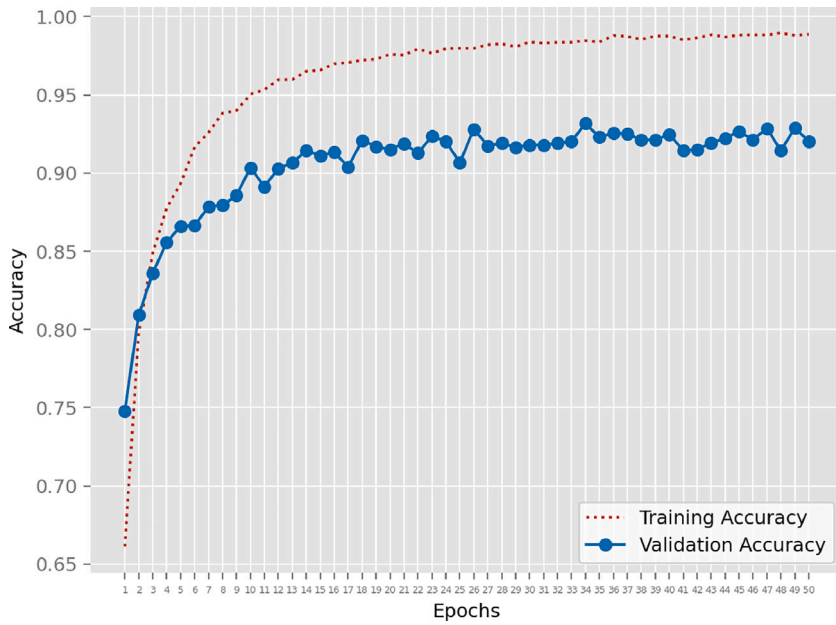

Fig. 3. Epoch-accuracy trend for the MobileNet model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

input image. In our case, it is worth noting that both comparison algorithms' results are restricted to a specific range. The SSIM range values range from −1 to +1 (often output as [0, 1]), while the FSIM ranges from 0 to 1, with 1 indicating that the images are identical. In general, higher values indicate greater similarity between heatmaps generated by different explainability methods, supporting model robustness and interpretability. These ranges allow us to compare different heatmaps based on absolute value and the same analysis criteria, and they provide useful information for comparing different DL models to one another.

4. Experimental analysis and results

In this section, we report the dataset under analysis with more details and all the results, both quantitative and qualitative, we obtained from the experimental analysis. In the final part, we discuss these results.

As mentioned before, the used dataset derives from the Kaggle website, at the following URL.[6] The dataset consists of 38.569 dermatoscopy images divided into seven classes. The applied splitting rate is the 80-10-10 for training, validation and testing folders, respectively, and this is shown in Table 2.

After the splitting phase, the dataset is trained and tested on the cited five CNNs. The hyper-parameters setting is Table 3.

This combination is found by empirically testing different combinations on all the considered networks. All training and tests were performed in a working environment with a laptop equipped with an Intel Core i7 CPU with 16 GB RAM.

In Table 4 are reported the metrics of the networks in terms of accuracy, precision, recall, F-Measure, and loss.

As shown in Table 4, the best performances are related to the MobileNet network with 0.93 in accuracy, precision, and recall, while the loss is 0.30. Also networks such as the ResNet50 and Standard_CNN report good performance. From these results, it is possible to deduce that this network (i.e. MobileNet) can correctly classify the seven skin lesion conditions.

Looking the Fig. 1 in Section 3, the best model is the MobileNet, so the next analysis regards this generated model with this performance.

[6] https://www.kaggle.com/datasets/artakusuma/basedir/data

Table 2
Dataset splitting.

Folder	Class	# Samples	Total	Splitting
Training	akiec	4.173	30.854	80
	bcc	4.686		
	bkl	4.736		
	df	3.528		
	mel	4.736		
	nv	4.763		
	vasc	4.232		
Validation	akiec	521	3.855	10
	bcc	585		
	bkl	592		
	df	441		
	mel	592		
	nv	595		
	vasc	529		
Testing	akiec	523	3.860	10
	bcc	587		
	bkl	592		
	df	441		
	mel	592		
	nv	596		
	vasc	529		

Table 3
Hyperparameters setting.

Epochs	Batch size	Learning rate	Image size
50	16	0.0001	224 × 3

To have a better overview of the trends during the training and validation phase, Figs. 3 and 4 show the plot of epoch accuracy and epoch loss, respectively.

Fig. 3 shows positive outcomes during the training phase and a minor decline during the validation phase. The accuracy plot illustrates that training accuracy (red dotted line) increases and plateaus, whereas validation accuracy (blue line) increases and stabilizes, or reduces if overfitting occurs. This is a predicted behavior, as both training and validation accuracy demonstrate that the MobileNet model was able to learn the differences between skin lesion images from different classes. Fig. 4 demonstrates the opposite behavior acquired from the (training and validation) loss, providing more evidence that the model

Table 4
Metrics evaluation for tested DL models for the HAM10000 dataset.

Networks	Accuracy	Precision	Recall	F-Measure	AUC	Loss
EfficientNet	0.154	0.154	0.154	0.154	0.503	13.752
MobileNet	0.936	0.937	0.935	0.936	0.986	0.309
ResNe50	0.902	0.904	0.900	0.902	0.975	0.540
Standard_CNN	0.845	0.846	0.846	0.846	0.976	0.493
InceptionV3	0.873	0.943	0.723	0.818	0.985	0.392

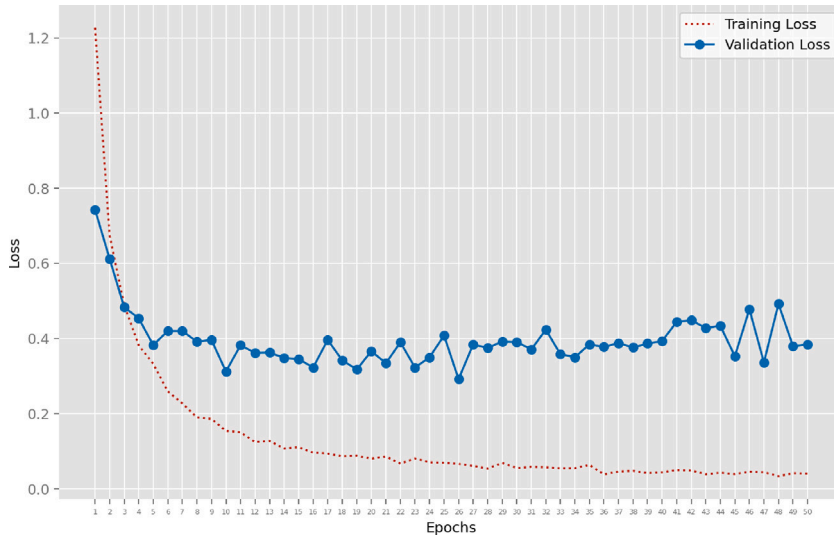

Fig. 4. Epoch-loss trend for the MobileNet model.

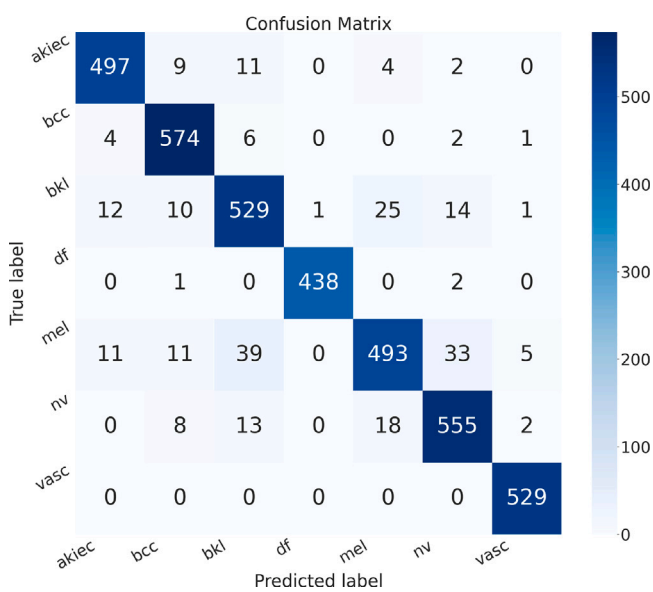

Fig. 5. The confusion matrix for the MobileNet model.

is accurately learning the distinctions between the seven classes under examination. Little validation accuracy and loss fluctuations throughout a CNN's training phase are common. They can be linked to several variables about the training procedure, the data's properties, and the model's behavior.

Moreover, to observe how the different classes are classified during the testing phase, Fig. 5 reports the confusion matrix for the MobileNet network. The confusion matrix, in Fig. 5 reveals the good performance of the model, with higher values on the first diagonal of the matrix, and this means that objects labeled in a certain class are correctly predicted in the correct class.

From the statistical point of view, the quantitative analysis is concluded with the application of the correlation matrix over the seven classes, considering as parameters the performances (accuracy, precision, recall, AUC and F-Measure) of the MobileNet, reported in Fig. 6

Fig. 6 shows the 7×7 correlation matrix generated using the metrics of the MobileNet related to each skin class. The cells contain numerical values representing the Pearson correlation coefficients between each pair of classes. The minimum and maximum values are -1 and $+1$, indicating no correlation and maximum correlation between the classes, respectively. The matrix highlights that the MobileNet model's performance on certain skin disease classes is not independent. Strong positive correlations, such as DF and MEL, suggest that improving the model for one class might concurrently improve it for the correlated class; and vice versa, negative values, like DF and BCC, behave relatively independently (or inversely) in terms of model performance. High positive correlation indicates that the MobileNet model likely finds these classes either similarly easy/difficult to classify, or they share common visual features that the model learns well. ON the other hand, negative correlation implies that the visual features are quite distinct, or that the model is processing them in fundamentally different ways.

Moving to the qualitative aspect, the best model, i.e. MobileNet, is evaluated by considering the visual explainability feature. With this aim, the authors applied the tested images to the CAM algorithms, shown in Figs. 7 and 8 the Grad-CAM and Score-CAM applications, respectively.

The Figs. 7 and 8 report for each of the seven classes, the input image (on the left), the generated heatmap (in the middle), and the overlap of the previous two (on the right). Some considerations are discussed below:

- The generated heatmaps correspond to the presence of the skin lesions in the image, meaning the classification was obtained according to the recurrent pattern (the skin lesion) in the images.

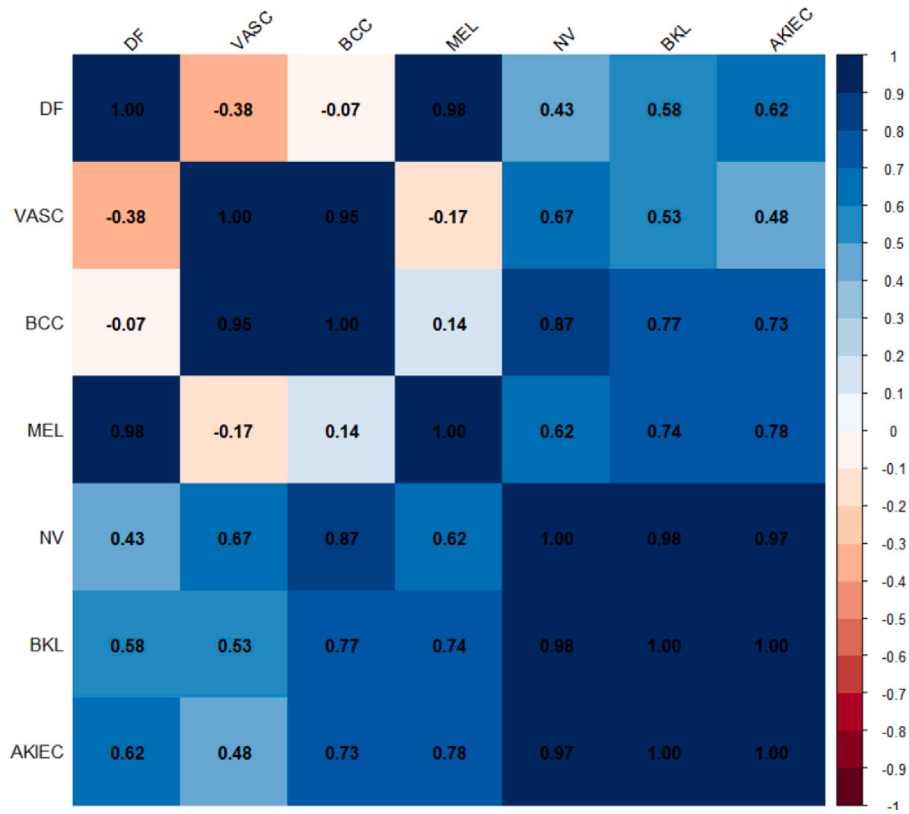

Fig. 6. The correlation matrix for the MobileNet model.

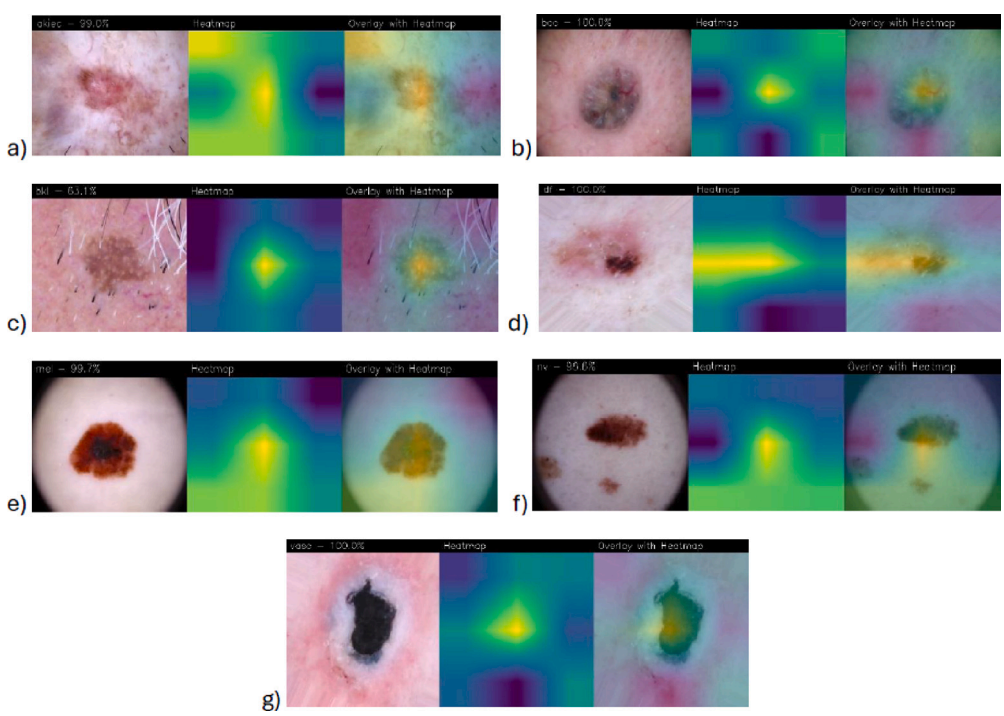

Fig. 7. Grad-CAM application on input images: (a) akiac, (b) bcc, (c) bkl, (d) df, (e) mel, (f) nv, (g) vasc.

- The high performances on accuracy and precision, mixed with the CAM algorithms guarantee a correct classification through the analyzed classes. So the model considers in the training process, several features of the images such as the shape, dimension, coloration, etc. of the skin lesions.
- In some cases, the areas highlighted from CAMs also include other areas such as the corner of the images or lower/higher areas.
- Another interesting consideration regards the similarity between the Grad-CAM and Score-CAM. These two algorithms work differently, but they recognize the same areas with slight differences,

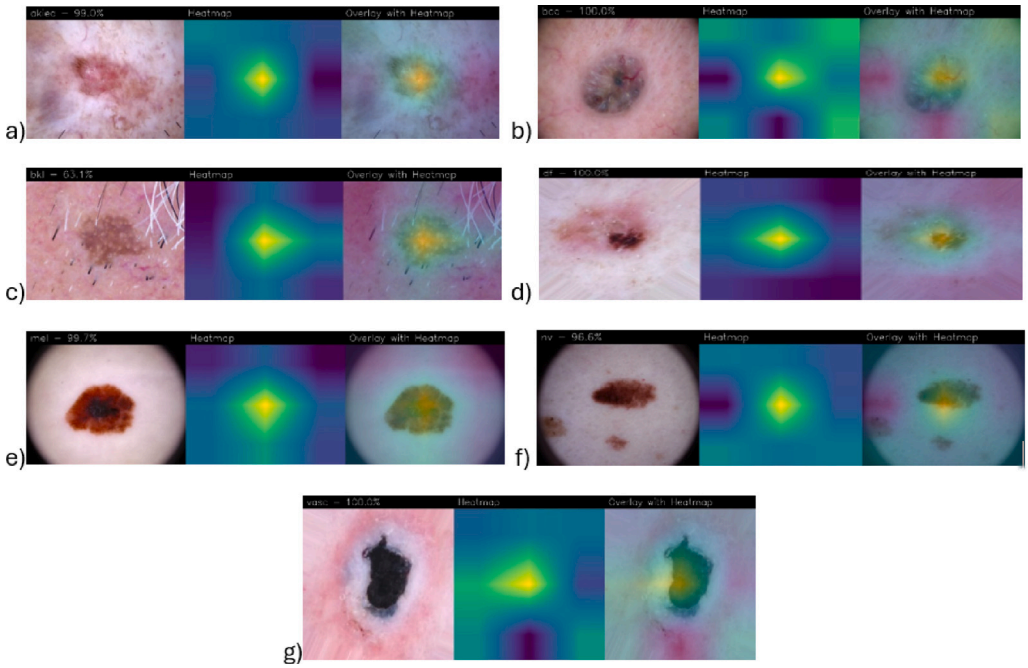

Fig. 8. Score-CAM application on input images: (a) akiac, (b) bcc, (c) bkl, (d) df, (e) mel, (f) nv, (g) vasc.

Table 5
Similarity indices for the MobileNet model.

Classes	FSIM Grad-CAM	FSIM Score-CAM	IM-SSIM
df	0.654	0.605	0.915
vasc	0.695	0.685	0.946
bcc	0.622	0.657	0.873
mel	0.635	0.724	0.844
nv	0.695	0.673	0.906
bkl	0.630	0.688	0.820
akiec	0.600	0.670	0.816

improving the robustness of the model and the trustworthiness of the medics.

Compared to SkinSage XAI's 96% accuracy, the MobileNet-based model provides a competitive 93.6% accuracy while offering a lightweight architecture with reduced computational complexity and high explainability consistency across both Grad-CAM and Score-CAM methods.

In addition, the authors propose a set of indices aimed to quantify the qualitative results and improve the model's robustness, by introducing the FSIM and SSIM. Table 5 shows the average similarity value between the two sets of Grad-CAM and Score-CAM heatmaps for each class.

As reported in Table 5, the FSIM evaluated from the Grad-CAM and Score-CAM folders is over 0.60. remembering that this measure compares all the heatmaps present in a single class, these results confirm the reliable choice of dataset, and on the other hand the good generalization for DL classification purposes. In other words, if these values were close to zero, that means the images are not similar (belonging to another class); otherwise, the high values close to 1 describe images too similar to each other and without an appropriate generalization. This behavior is repeated for both Grad-CAM and Score-CAM heatmaps.

Regarding the SSIM, this index compares the heatmaps of each class generated with the Grad-CAM with the corresponding one generated with the Score-CAM. As expected, Table 5 shows high values (> 0.80) of SSIM, guaranteeing that the two CAM algorithms reveal the same areas with little differences due to the different operating principles of the algorithms. From the medical point of view, these analyses represent a method to identify and locate skin lesions in patients, improving the credibility of this technology, and turning the "black box AI approach into an explainable AI one.

4.1. Experimental analysys #2

As mentioned in the Section 1, in this subsection, we provide further tests on the networks to improve the reliability and robustness to validate the previous quantitative results. A new dataset is taken in account. This dataset is provided by Prasad Maharana from the Kaggle website at the following link[7]: This dataset is composed by eight classes: the seven already analyzed and the Squamous cell carcinoma (SCC). This last class is eliminated to maintain coherence with the previous dataset. Also, the number of images for each class is reduced to the same number for each to avoid unbalanced problems. The final number of samples is 2.100 divided into the seven classes and split in a 80-10-10 rate for training, validation and testing. The results, considering the same networks and the same hyper-parameters setting, are reported in Table 6.

These results confirm the same considerations made with the previous dataset. MobileNet and InceptionV3 report good performances with 80.6% and 83.8% accuracy, respectively. ResNet50 and StandardCNN show a similar behavior with a decrease of performance, which stands at 68.9% and 68.0%, respectively. Table 6 also confirms the results of EfficientNet, with a significant problem in recognizing retinal images between classes. Finally, the general reduction of the performance of the CNNs is associated with the small number of images for each class; however, the performance behavior of the CNNs reports the same metrics trends, given to the further analysis (CAMs and SSIM) a higher value of reliability and credibility.

5. Conclusion and future work

The research paper focused on utilizing DL models for seven-class skin lesion classification, aiming to create a reliable and accurate

[7] https://www.kaggle.com/datasets/salviohexia/isic-2019-skin-lesion-images-for-classification?select=MEL.

Table 6
Metrics evaluation for tested DL models on Prasad Maharana's dataset (balanced classes, second validation set).

Networks	Accuracy	Precision	Recall	F-Measure	AUC	Loss
EfficientNet	0.157	0.163	0.123	0.140	0.553	3.345
MobileNet	0.804	0.808	0.804	0.806	0.918	2.248
ResNe50	0.685	0.697	0.680	0.689	0.892	1.726
Standard_CNN	0.680	0.696	0.666	0.681	0.915	1.263
InceptionV3	0.838	0.859	0.850	0.843	0.959	1.297

classification, i.e. MobileNet with 0.935 in accuracy, system while providing an understandable logic behind the model. The study successfully demonstrated the effectiveness of the CAM algorithms in correctly classifying skin lesions through the analyzed recurrent pattern in the classes. The research not only improved the model's robustness but also quantified qualitative results by introducing FSIM and SSIM metrics, further enhancing the accuracy and precision of the classification process. Furthermore, the SSIM values exceeding 0.80 between the heatmaps generated by Grad-CAM and Score-CAM highlighted the consistency in identifying relevant areas of skin lesions, emphasizing the potential of the model for accurate lesion localization and diagnosis. Overall, the study successfully developed a transparent and trustworthy tool that aids dermatologists in diagnosing skin lesions accurately, advancing the prevention and early intervention of skin cancer.

Future research will investigate the use of Transformer-based models to improve the accuracy and robustness of skin lesion detection and localization, exploiting their excellent ability to capture long-range correlations inside medical images. Regarding the limitation of the fair skin tones, future work will include augmenting the dataset with diverse skin tones or evaluating performance on more demographically representative datasets. Additionally, attempts will be made to optimize the created models for real-time inference on edge devices, allowing for speedy and accessible diagnostic support in resource-constrained contexts. To evaluate the therapeutic efficacy and social impact of our research, further steps will include clinical trials and real-world deployment scenarios, ensuring that our explainable AI architecture is seamlessly integrated into dermatological practice.

CRediT authorship contribution statement

Marcello Di Giammarco: Writing – original draft, Visualization, Methodology, Data curation. **Antonella Santone:** Supervision. **Mario Cesarelli:** Supervision. **Fabio Martinelli:** Supervision. **Francesco Mercaldo:** Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Dataset link: https://www.kaggle.com/datasets/artakusuma/basedir/data.

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